



Assignment Brief (RQF)

Higher National Diploma in Computing

Unit 7: Software Development Lifecycles

Assignment Brief

Student Name/ID Number	
Unit Number and Title	Unit 7: Software Development Lifecycles
Academic Year	2026
Unit Tutor	Ms. Yee Mon
Assignment Title	Assignment 2 of 2: Software Development Lifecycles and Software Behavioural Design Techniques Analysis
Issue Date	25/5/2026 (Monday)
Submission Date	27/6/2026 (Saturday)
Submission Format	
The submission is in the form of an individual written report and presentation. You should submit the report in pdf and presentation file in pdf . This should be written in a concise, formal business style using single spacing and font size 12. You are required to make use of headings, paragraphs and subsections as appropriate, and all work must	

be supported with research and referenced using the Harvard referencing system. Please also provide a bibliography using the Harvard referencing system.

- Use Microsoft Word for documentation and report.
- Your report must be formatted as follows:

➤	Font Name	:	Verdana
➤	Font Size	:	.12 pt
➤	Margin	:	75 inches (round)
➤	Line Spacing	:	1.5 pt
➤	Alignment	:	Justify
➤	Left Hand Side of Header	:	Batch Number
➤	Right Hand Side of Header	:	Your Name
➤	Centre of Footer	:	Page Number
- Use the Harvard referencing system.
- Complete the title page and sign the statement of authenticity.
- Your assignment must be submitted in pdf format.

Unit Learning Outcomes

The purpose of this assignment is to:

LO3 Undertake a software development lifecycle

LO4 Discuss the suitability of software behavioural design techniques

Transferable skills and competencies developed

On successful completion of this unit, students understand undertaking the software development lifecycle (SDLC) equips students with both technical and transferable skills that are highly applicable across multiple domains and behavioural design techniques that focus on how users interact with software, and how the system should behave in response to various inputs.

Vocational scenario

You are a junior software development team working at a local IT company in Myanmar.

A university is facing transportation problems. Students complain that campus shuttle buses are:

- ❖ Often late
- ❖ Overcrowded
- ❖ Difficult to track

The university management wants to develop a computerized **Smart Campus Shuttle System**. You are a junior trainee software developer in the IT company. The university management expects to implement a computerized **Campus Shuttle System** that can fulfil the following activities such as

- ❖ Track buses in real-time
- ❖ Show arrival times via a mobile app
- ❖ Allow students to report issues

Assignment activity and guidance

As part of the project team, you've been tasked with writing a project briefing report for your manager that includes the following facts such as SDLC, Risk Management in software models and Feasibility Report Discussion—

Task-3: Undertake a Software Development Lifecycle

Undertake a software development lifecycle for the development of the above web application to produce **requirements specification** and **design documents (use case, class diagram, sequence diagram, activity diagram, etc.,)**. In your development, you should include the following:

- Explore software investigation to meet a business need. Your investigation should identify a business need and propose a

software solution to meet the requirements of the client.

- Use appropriate software analysis tools/techniques to carry out the investigation and create supporting documentation, including use cases, requirements specifications, and design documents to support your solution.

Furthermore, you should analyse how software requirements can be traced throughout the software lifecycle. And you should also discuss two approaches to improving software quality as well as critically evaluate the process of undertaking a systems investigation regarding its effectiveness in improving a software quality.

Task-4: The Suitability of Software Behavioural Design Techniques

Write a report on the suitability of software behavioral design techniques for the given project. In your report, you should address the following points:

- Define software behavioral design techniques and their importance in software development.
- Discuss the suitability of different software behavioral design techniques for the given project, including examples.
- Evaluate the benefits and limitations of using software behavioral design techniques for this project.

Finally, you should analyse a range of software behavioural tools and techniques and differentiate between a finite state machine (FSM) and an extended FSM, providing an application of use for both. Additionally, you should make justifications of how data-driven software can improve the reliability and effectiveness of software for successful software development projects.

In your presentation slides, you should prepare according to the following facts such as:

1. What is the SDLC and Stages in SDLC
2. What are the key Activities in each SDLC phase
3. What are software analysis tools / techniques with example
4. Introduction to Behavioural Design Techniques
5. Role of Behavioural Design in SDLC with example
6. Evaluating Suitability of Behavioural Design Techniques
7. Benefits and Limitations

Submission Guidelines:

You need to submit **two** files:

Presentation **slides** in a pdf format compatible with common presentation tools.

Pdf file format **Report**: Submit written reports in a standard document format with proper formatting and citation.

This assignment is designed to assess your ability to apply software development concepts in real-world scenario. Tailor your recommendations to the unique requirements and objectives of **Campus Shuttle System** Project.

Note to Students:

- ☐ Check carefully the submission date and the instructions given with the assignments
- ☐ Late assignments will not be accepted, except for valid circumstances with supporting evidence.
- ☐ Ensure that you give yourself enough time to complete the assignment by the due date.

- ☐ Do not leave things such as printing to the late minute-excuses of this nature will not be accepted for failure to hand-in the work on time.
- ☐ You must take responsibility for managing your own time effectively.
- ☐ If you are unable to hand in your assignment on time and have valid reasons such as illness, you may apply (in writing) for an extension.
- ☐ Take great care that if you use other people's work or ideas in your assignment, you properly reference them in your text and any bibliography.

Note: Plagiarism is an offence. Plagiarized work will be referred

Interim Review Session (Formative Assessment)

- ☐ 8/6/2026 (Monday)
- ☐ 9/6/2026 (Tuesday)

Submit the work to moodle.gusto-education.com by

- ☐ 11: 55 PM on 8/27/2026 (Saturday)

Sources of information:

You must provide of your references, internet sites or book names. Use Harvard referencing for book references.

Pass	Merit	Distinction
L03 Undertake a software development lifecycle		D3 Evaluate the process of undertaking a systems investigation with regard to its effectiveness in improving a software quality.
P5 Undertake a software investigation to meet a business need. P6 Use appropriate software analysis tools/techniques to carry out a software investigation and create supporting documentation.	M3 Analyse how software requirements can be traced throughout the software lifecycle. M4 Discuss two approaches to improving software quality.	
L04 Discuss the suitability of software behavioural design techniques		D4 Present justifications of how data-driven software can improve the reliability and effectiveness of software.
P7 Discuss, using examples, the suitability of software behavioural design techniques.	M5 Analyse a range of software behavioural tools and techniques. M6 Differentiate between a finite state machine (FSM) and an extended FSM, providing an application of use for both.	